Project Name: Swag Labs (SauceDemo)
Application URL: https://www.saucedemo.com
Testing Type: Functional, UI, Negative, Exploratory

1. Introduction

The purpose of this test plan is to verify the functionality, usability, reliability and overall quality of the Swag Labs e-commerce website.

Swag Labs is an online shopping application where users can log in, browse products, add products to their shopping cart, complete checkout and receive order confirmation.

The application will be tested from an end-user perspective to identify functional defects, UI issues, validation problems and unexpected application behaviour.

2. Testing Objectives

- Verify that users can log in with valid credentials.
- Verify proper error messages for invalid and locked-out users.
- Verify product details, images, descriptions and prices.
- Verify add-to-cart and remove-from-cart functionality.
- Verify shopping cart item count and product details.
- Verify checkout form validations.
- Verify subtotal, tax and final total calculations.
- Verify successful order completion.
- Identify hidden bugs through exploratory testing.
- Verify application compatibility across different browsers.

4. Types of Testing

Testing type	**Purpose**
Functional Testing	Verify application functionality
UI Testing	Verify layout, buttons and images
Negative Testing	Verify invalid input handling
Exploratory Testing	Discover unexpected defects
Smoke Testing	Verify critical application functionality

End-to-End Testing	Verify complete shopping journey

5. Test Environment

Environment	Configuration
Operating System	Windows 10/11
Browser 1	Google Chrome
Browser 2	Microsoft Edge
Testing Tool	Manual Testing
Automation Tool	Selenium WebDriver
Programming Language	Java
Build Tool	Maven
Test Framework	TestNG

These are the planned test environments; record actual versions and devices used during execution.

6. Test Data

Username	Password	Purpose
standard_user	secret_sauce	Normal functionality
locked_out_user	secret_sauce	Login restriction
problem_user	secret_sauce	Identify intentional bugs
performance_glitch_user	secret_sauce	Performance observation

Additional test data:

- Invalid username: invalid_user
- Invalid password: wrong123
- First Name: Abhi
- Last Name: Patil
- Postal Code: 416416

7. Overall Test Scenarios

8. Automation Testing Strategy

Automation will be performed using Selenium WebDriver with Java and TestNG.

Critical automated test flows:

1. Login using standard_user.
2. Add product to cart and complete checkout.
3. Login using locked_out_user and verify the error message.

Automation best practices:

- Use stable element locators.
- Use explicit waits.
- Add assertions for expected results.
- Maintain reusable test methods.
- Capture screenshots on failure.
- Generate execution reports.

9. Entry and Exit Criteria

Entry criteria

- Application is accessible.
- Test accounts and credentials are available.
- Test environment is configured.
- Test cases are prepared.

Exit criteria

- All critical test cases are executed.
- Failed test cases are documented.
- Identified defects are reported.
- Critical defects are communicated.
- Automation execution results are documented.
- Test execution summary is prepared.

10. Final Deliverables

Deliverable	Description
Test Plan	Overall testing strategy
Test Cases	Detailed manual test cases
Bug Report	Identified and documented defects
Automation Code	Selenium automation scripts
README	Automation execution instructions
Video 1	Test plan and automation demonstration
Video 2	Manual bug discovery demonstration

11. Conclusion

The objective of this QA assignment is to evaluate the Swag Labs application through structured manual and automation testing. Testing will focus on the complete e-commerce workflow, including login, product selection, shopping cart, checkout and order confirmation.

By applying functional, negative, exploratory, UI and regression testing techniques, the QA process aims to identify defects, improve application reliability and ensure a smooth user experience.

Important before submission: Your existing bug report and manual test workbook should be checked for duplicate test IDs and incorrect expected results. In particular, the locked-out user behaviour is expected functionality, not a defect.

You can use this as your overall test-plan content and transfer it into Word or PDF. Make sure you personally validate the scenarios and follow the assignment's instruction about independent work.